BLOCKCHAIN TECHNOLOGY (CSE4444)

ASSIGNMENT 2 — DECENTRALIZED APPLICATION (DAPP) DEVELOPMENT

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Assignment Title	Blockchain Technology (CSE4444) — Assignment 2
Title of Developed DApp	**MediTrace: Pharmaceutical Authenticity & Batch Recall Registry**
GitHub Repository Link	**https://github.com/kristanuxuzir/MediTrace-DApp**

1. Project Overview & Problem Addressed

According to the World Health Organization (WHO), counterfeit and substandard medicines compromise over 10% of medical supplies in developing nations, leading to treatment failures, drug resistance, and avoidable fatalities. Traditional centralized pharmaceutical registries are prone to unauthorized data manipulation and suffer from prolonged delays when recalling contaminated drug batches.

MediTrace solves this real-world problem by deploying an immutable Ethereum smart contract registry. Health regulators license accredited pharmaceutical manufacturers on-chain. Manufacturers register production batches with unique batch IDs, expiry timestamps, unit counts, and cryptographic quality assurance certificate hashes. In case of contamination, instant recalls can be triggered, alerting pharmacies and patients globally in real time.

2. Key Components & Implementation Details

• **Smart Contract (contracts/MediTrace.sol):** Implemented in Solidity 0.8.20 with role-based access control (onlyOwner, onlyManufacturer), gas-efficient custom errors, events, and zero-gas public verification.

• **Web Frontend (frontend/):** Built using separate HTML5, CSS3, and JavaScript files featuring 4 dedicated user portals: Public Verification, Manufacturer Portal, Regulatory Authority, and Live Ledger.

• **MetaMask Integration (Ethers.js v6):** Connects wallet, detects active network (Sepolia/Ganache) and chain ID, dynamically determines user permissions, sends signed state-changing transactions, and queries view calls.

• **Automated Testing (test/MediTrace.test.js):** 27 automated Hardhat unit tests achieving a 100% pass rate across valid, invalid, boundary, and unauthorized input conditions.

• **Complete Documentation (README.md):** Includes full Remix IDE compilation and deployment guides, test execution matrix, 6-item screenshot checklist, and technical viva voce questions.

3. Evaluator Access & Integrity Declaration

The complete source code, deployment instructions, test cases, and frontend assets are organized in the GitHub repository at https://github.com/kristanuxuzir/MediTrace-DApp. No private keys, credentials, or wallet seed phrases are stored or uploaded.